

A cyclodextrin polymer as supramolecular matrix for scalable green photooxygenation of hydrophobic substrates in homogeneous phase

Marco Agnes^{a*}, Arianna Mazza^a, Eszter Kalydi^b, Szabolcs Béni^b,
Milo Malanga^c and Ilse G. J. Manet^{a*}

a. Institute for Organic Synthesis and Photoreactivity (ISOF), National Research Council (CNR), via P. Gobetti 101, 40129 Bologna, Italy

b. Department of Pharmacognosy, Semmelweis University, Üllői út 26., 1085 Budapest, Hungary

c. CarboHyde Zrt., Berlini str., 47-49. Budapest, 1045 – Hungary

Corresponding authors:

Ilse Manet, e-mail: ilse.manet@isof.cnr.it

Marco Agnes, e-mail: marco.agnes@isof.cnr.it

Table of contents graphic:



Aromatic endoperoxides were synthesized as O₂ releasing agents in organic solvent upon irradiation of a photocatalyst in the presence of the precursors in mM concentration. Co-encapsulation of both reactants in a cyclodextrin polymer allowed to successfully perform the photocatalyzed oxygenation in homogeneous aqueous environment. Controlled release of O₂ could be achieved upon thermolysis.

Introduction

Nowadays, the design and development of biocompatible molecular vectors able to deliver molecular oxygen (O_2) in tissues is gaining significant attention in the pharmacological field thanks to their application potential in the reversion of hypoxia.^[1] Hypoxic conditions are characterized by low O_2 tension, typically below 2%, much less compared to healthy physiological amounts (normoxic environment) where O_2 tension is between 5-9%.^[2] This environment is typical of several diseases characterized by poor vascularization or impaired blood flow including solid tumors,^[3,4] microbial^[5] and fungal^[6] infections and biofilms, cerebral^[7] and myocardial^[8] ischemia, etc. These pathological conditions are among the most common death causes in populations where average age increases. Concerning the treatment of tumors and infections, low O_2 levels affect the efficacy of current drug-based therapies as it can induce drug resistance and, moreover, reduce the positive effects of radiotherapy^[9] and photodynamic therapy (PDT),^[1] both relying on normoxic conditions to produce, upon suitable irradiation, reactive oxygen species (ROS) like singlet oxygen (1O_2). In this context, a large amount of research capitalized on aromatic endoperoxides as suitable Oxygen-Releasing Agents (ORAs). Polycyclic aromatic hydrocarbons are cheap and versatile substrates able to reversibly bind and release O_2 upon suitable triggering (**Figure 1A**).^[10] The endoperoxides obtained after reaction with 1O_2 have been widely studied for several decades,^[11] along which many insights regarding their preparation and reactivity have been obtained.^[12] However, an approach towards the optimization of the synthesis envisaging industrial scale up of a sustainable and circular production process is still lacking.

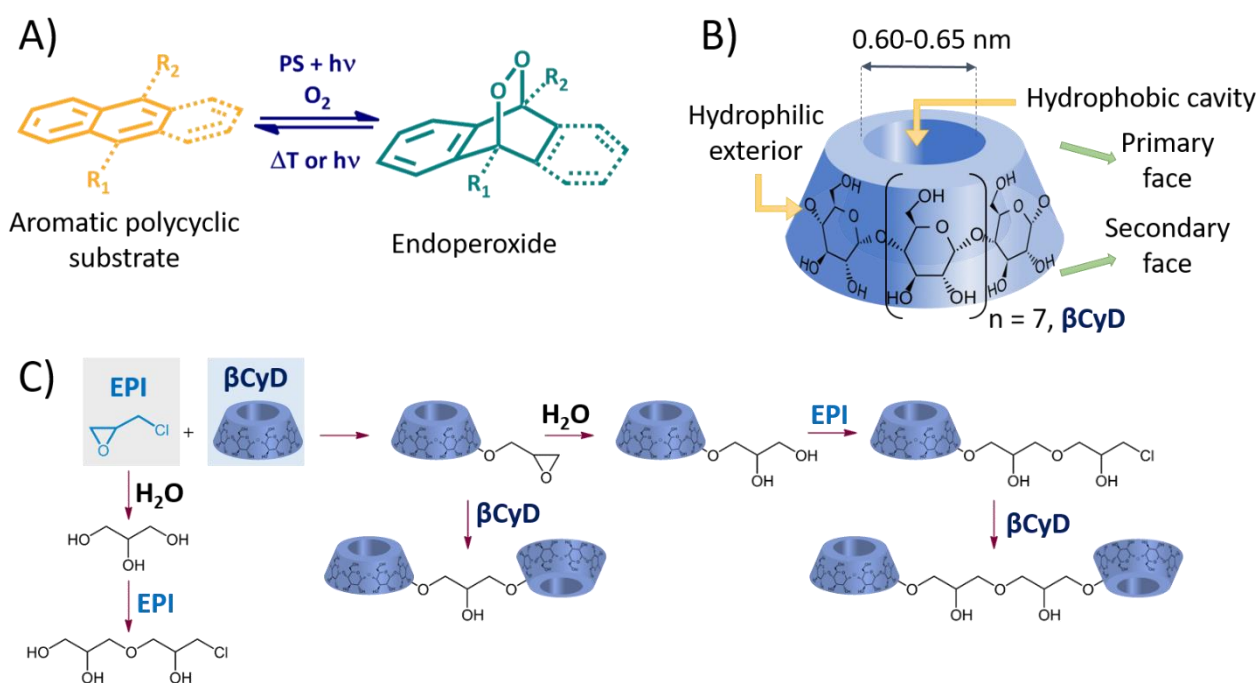


Figure 1: A) General scheme of the reversible endoperoxide formation of aromatic substrates; B) Chemical structure of β -cyclodextrin (β CyD); C) Scheme of the polymerization of β CyD with epichlorohydrin (EPI) to form BCDPS.

The group of W. Fudickar and T. Linker developed a comprehensive library of anthracene^[13,14] and naphthalene^[15,16] endoperoxides investigated as ORAs. In most cases, these compounds showed very poor

aqueous solubility, thus the oxidation of the aromatic substrates was achieved in organic solvent by photosensitized $^1\text{O}_2$ produced upon red light irradiation in the presence of a suitable photosensitizer (PS). Careful selection of the aromatic substrates and the reaction conditions can afford a single photoproduct incorporating the target O-O bridge. Moreover, suitable derivatization of the aromatic core showed implications on their solubility and, above all, on their ability to selectively release O_2 . Rather than spontaneous return to the parent molecule, a trigger is preferred to release O_2 from the ORAs and, in this frame, thermolysis is the best performing strategy,^[17] even though also photolysis or chemically-triggered release of O_2 have been achieved.^[18,19]

In order to tackle the challenge of producing ORAs from aromatic substrates embracing an approach based on green chemistry paving the way for future scale up, cyclodextrin (CyD) chemistry was exploited. CyDs consist of α -D-glucopyranose units joined by $\alpha(1-4)$ linkages forming macrocyclic oligosaccharides with a hydrophilic external surface and a hydrophobic cavity suitable to encapsulate hydrophobic guests.^[20] Most commonly used biocompatible monomers are α CyD, β CyD and γ CyD comprising 6, 7 or 8 glucose units, respectively (**Figure 1B**). Monomeric CyDs are known to work as reaction vessels,^[21] acting as catalyst^[22] in both heterogeneous^[23] and homogeneous^[24] conditions. Actually, β CyD-derivatives have been described as water-solubilizing agents enabling the photooxygenation of organic molecules in aqueous environment: i) the supramolecular inclusion of a porphyrin into permethylated β CyD-monomer allowed to oxidize L-methionine methyl ester and uracil with high turnover numbers and minor photobleaching of the PS;^[25] ii) Zn(II) phthalocyanines covalently linked to permethylated β CyD-monomer catalysed the photooxygenation of 1-naphthol and 2-furoic acid with high conversion efficiency and reaction yields, displaying higher photostability and recyclability of the PS.^[26] Besides, native and 2-hydroxypropylated CyD-monomers showed interaction with anthracene-endoperoxides, with a stabilizing effect on the O-O bridge.^[27]

Also polymeric networks based on CyDs have shown interesting ability to catalyse in heterogeneous phase the hydrogenation of aromatic molecules^[28] or the formation of metal-nanoparticles,^[29] but homogeneous phase catalysis involving CyD polymers as reaction matrix has not been reported so far. These biocompatible polymers^[30] are fascinating materials mainly for three factors:^[20,30] i) straightforward synthesis in water (polycondensation of CyD monomers with a crosslinking agent, *i.e.* epichlorohydrin, BCDPS, **Figure 1C**); ii) improved aqueous solubility (β CyD monomers reach ≈ 0.01 M, while in BCDPS of MW ≈ 50 kDa and β CyD content $\approx 70\%$ w/w, the $[\beta\text{CyD}]$ is ≥ 0.10 M); iii) ability to encapsulate more guests simultaneously.

The aim of this work was to investigate BCDPS as supramolecular reaction vessel, focusing on the optimization of the photosensitized oxygenation of aromatic molecules in homogeneous aqueous environment and the preliminary observation of the O_2 release from the endoperoxide substrates. UV-Vis and NMR spectroscopies were used as orthogonal analytical techniques to monitor both reactions. The obtained data confirmed the possibility to achieve the photooxygenation of hydrophobic substrates in non-deuterated water upon complexation with the polymer. Importantly, the protocol required three readily

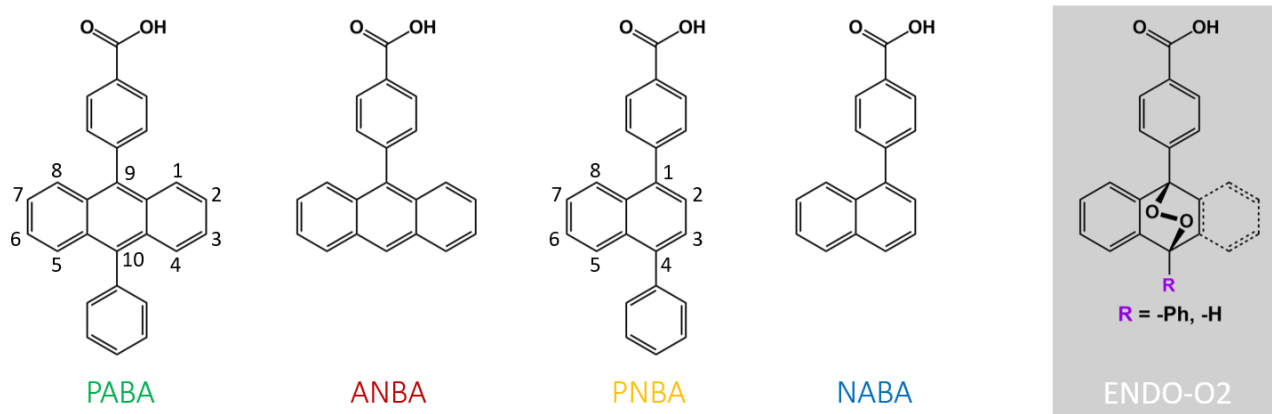
accessible reagents dissolved together in buffered water, BCDPS, the PS and the aromatic substrates, all of them produced on large scales at affordable costs. Moreover, the polymer could be recovered after the photooxygenation making the protocol fit with the principles of circular chemistry. Finally, a proof-of-concept was provided of a scalable green reaction protocol exploiting for the first time the CyD polymer matrix to enable homogeneous photocatalytic reactions. Thanks to the CyD ability to include and dissolve various hydrophobic molecules, this process can be easily adopted for the reaction of other hydrophobic substrates.

Results and Discussion

Experimental design: selection of target molecules

The requirements for an ideal chemical source of O₂ are the straightforward generation of the O₂-carrying form and its high propensity to release O₂ upon selective triggering. 9,10-Disubstituted anthracenes are known to form stable endoperoxides photochemically rather smoothly, while naphthalene derivatives are less prone to react with photosensitized ¹O₂ but their endoperoxides are less stable under environmental conditions and relative O₂ release is faster.^[10]

In this context, two pairs of mono- and di-substituted anthracene (**PABA**, **ANBA**) and naphthalene (**PNBA**, **NABA**) derivatives bearing one or two aromatic substituents were synthesized and tested for photooxygenation in order to generate their endoperoxide analogues (ENDO-O₂) (**Figure 2**). A carboxylic function has been inserted in order to increase the aqueous solubility of the molecules, but also to offer a versatile moiety for further modification.

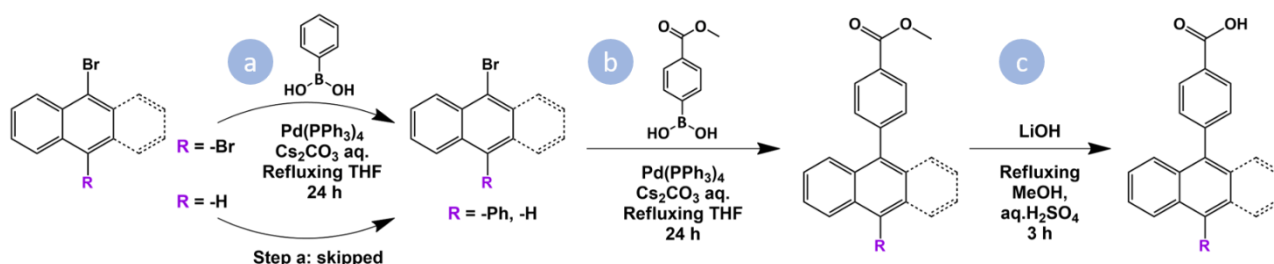


*Figure 2: Chemical structures of the synthesized aromatic molecules [4-(10-phenylanthracen-9-yl)benzoic acid, **PABA**, 4-(anthracen-9-yl)benzoic acid, **ANBA**, 4-(4-phenylnaphthalen-1-yl)benzoic acid, **PNBA**, and 4-(naphthalen-1-yl)benzoic acid, **NABA**] with numbering of the aromatic cores and general structure of their endoperoxide analogues (ENDO-O₂).*

While the synthesis of **PNBA** has not been published before, **NABA**,^[31] **ANBA**,^[32] and **PABA**^[33] are known compounds. The latter has been taken as reference since its photooxygenation in organic environment and further covalent linkage to a polymeric matrix afforded preliminary satisfactory data for the photothermally-induced release of O₂ from the corresponding endoperoxide **PABA-O₂**. At the moment, there are no data on the photooxygenation of naphthalene-cores bearing aromatic substituents in the 1- and 4- positions.

Thermal synthesis of anthracene and naphthalene substrates

A previously published synthesis of **PABA**^[33] was adapted for the preparation of the whole set of materials. The optimized synthetic steps are shown in **Scheme 1** while the yields are reported in **Table 1**. A detailed description of the synthetic protocols for all the compounds is present in the **ESI** together with their full characterization achieved by ¹H and ¹³C NMR spectroscopy. The final products **PABA**, **ANBA**, **PNBA** and **NABA** were further analysed by ESI-MS (**ESI**).



Scheme 1: Synthetic steps for the preparation of the four aromatic derivatives (2 anthracenes, **PABA** and **ANBA**, and 2 naphthalenes, **PNBA** and **NABA**). Yields are reported in Table 1.

Table 1: Summary of the yields obtained during the synthesis of the four aromatic substrates **PABA**, **ANBA**, **PNBA** and **NABA** compared to the reported ones.

Compound	Step a (%)	Step b (%)	Step c (%)	Overall yield (%) Steps a-b-c	Overall yield (%) Steps b-c	Procedure
PABA	35	53	65	12	34	Ref. 31
	57	81	88	41	71	This work
ANBA	-	-	-	-	38	Ref. 30 (one step)
	-	58	77	-	45	This work
NABA	-	-	-	-	59	Ref. 29 (one step)
	-	85	65	-	55	This work
PNBA	68	83	75	42	62	This work

The Suzuki couplings of steps a and b were optimized following previously published procedures.^[34] The reactions were performed in THF instead of toluene/ethanol mixture and Cs₂CO₃ was used as base instead of Na₂CO₃. These changes allowed increasing the yields significantly (**Table 1**). The final deprotection step c was performed accordingly to the published procedure of **PABA**,^[33] but isolation of the final products could not be achieved by direct precipitation and it required extraction of the protonated compounds and collection after solvent evaporation. The overall yields of the three steps (a-b-c) increased 3.4 times (41% vs 12%) for **PABA** and similar yields could be obtained for **PNBA**. The synthesis of **ANBA** was published as a one-step reaction yielding 38% of the product.^[32] In this work, the two-step (b-c) procedure allowed to achieve an overall 45% yield. Similarly for **NABA**,^[31] while the reported one-step synthesis achieved a 59% yield, the two-step (b-c) synthesis resulted in an overall 55% yield.

Several publications are available on the photochemical preparation of endoperoxides from anthracene and naphthalene derivatives.^[10,11] The reaction follows a [4 + 2] cycloaddition mechanism between the lowest excited electronic state of molecular oxygen, $^1\Delta_g$ (herein referred as 1O_2), and the ground state of the polycyclic aromatic substrates. Selective irradiation of a PS absorbing visible light is used to produce 1O_2 , precluding direct substrate excitation potentially leading to the formation of byproducts.

One of the most explored PS for the reaction is methylene blue (MB) since: i) it is inexpensive with a good 1O_2 quantum yield (0.60 in water, 0.57 in DCM);^[35] ii) it is soluble in a wide variety of solvents including water (**Table S1**) where it aggregates at concentrations above 10 μM ;^[36,37] iii) it has a visible absorption band peaking around 660 nm, with a molar absorption coefficient of (ϵ) $\approx 80\,000\text{ M}^{-1}\text{ cm}^{-1}$ in organic solvent and $\approx 50\,000\text{ M}^{-1}\text{ cm}^{-1}$ in aqueous solutions (**Figure S1**). The PS aggregation in the solvent can be monitored from the *ratio* between the two absorption bands at 600 nm and 660 nm (**Table S2**). All together these features make MB a very attractive PS and it was selected for this study since it was also used in the published synthesis of **PABA-O2**.^[33]

First of all, the endoperoxide synthesis in organic solvent was optimized and **Table 2** collects the improved photooxygenation conditions compared to the previously published procedure. Reactions were performed in a mixture of dichloromethane (DCM) and tetrahydrofuran (THF), as pure DCM does not adequately dissolve the aromatic compounds in the mM range (**Table S1**) and THF does not dissolve MB. Being THF quite unstable under visible light irradiation,^[38] its amount was decreased and the final working mixture was DCM:THF 9:1 (v/v). The maximum concentration of dissolved O_2 in the solvent mixture at atmospheric pressure was assumed to be $\approx 2\text{ mM}$, according to the reported ones in pure DCM and THF (**Table S1**). Additional flow of gaseous O_2 was avoided because it needs careful handling due to its combustion hazard, especially in the presence of plausible fuels, *i.e.* endoperoxides. This feature has particular relevance in view of industrial scale up production. The used concentration of the aromatic substrates was 1 mM, allowing direct spectroscopic monitoring of the reaction having ϵ values between 10 000 and 18 000 $\text{M}^{-1}\text{ cm}^{-1}$ (**Figure S2**). The amount of MB was lowered to 50 μM (5% n/n with respect to the aromatic reagents).

Table 2: Optimized conditions for the photochemical step to obtain **PABA-O2** and **ANBA-O2**.

Proce- dure	[ORA]	[MB]	Solvent	Batch (mL)	Lamp	Optical filter	t (h)	T (°C)	Gas inlet	Yield
Ref. ^[33]	9 mM	0.9 mM (10%)	DCM 2 THF 1	15	Hg arc 400 W	White light	3	-78	O ₂	55% PABA-O2
Opt.	1 mM	0.05 mM (5%)	DCM 9 THF 1	20	Hg arc 100 W	577 nm (band $\pm$ 10 nm)	≤ 1	25	None (atm. press.)	94% PABA-O2 84% ANBA-O2

Possible ground and excited states interactions of MB with the aromatic substrates were checked under the selected conditions, as they may compromise the production of $^1\text{O}_2$ by MB. The absorption spectra of the reaction mixtures are similar but not identical to the sum of the spectra of equimolar solutions of the single components, indicating the occurrence of a ground state interaction altering the electronic properties (**Figure S3**). However, MB fluorescence properties were evaluated with and without the aromatic substrates and the fluorescence intensity changes were negligible ($< 2\%$, **Figure 3A**). Accordingly, the fluorescence lifetime (τ) remains constant in the samples ($\tau_1 \approx 0.2$ ns and $\tau_2 \approx 1.5$ ns, **Table S3A**) suggesting that neither static nor dynamic quenching processes of the MB excited state seem to occur under the experimental conditions chosen. Moreover, comparison of the fluorescence spectra of the aromatic substrates with and without MB, (**Figure S4**), falling in a range where MB does not absorb, is not easy due to MB absorption at the excitation wavelength, but their fluorescence lifetimes remain constant (**Table S3B**).

Eventually, direct measurement of the $^1\text{O}_2$ phosphorescence produced by MB was performed (**Figure 3B**). After exciting the samples at 650 nm, the $^1\text{O}_2$ production yield of MB decreased $\approx 12\%$ in the presence of three ORAs and only in case of **PNBA** the yield was reduced $\approx 50\%$. All together, these data indicate that in spite of the ground state interactions between aromatic molecules and MB, the photochemical behaviour of the PS was not compromised under the photooxygenation reaction conditions.

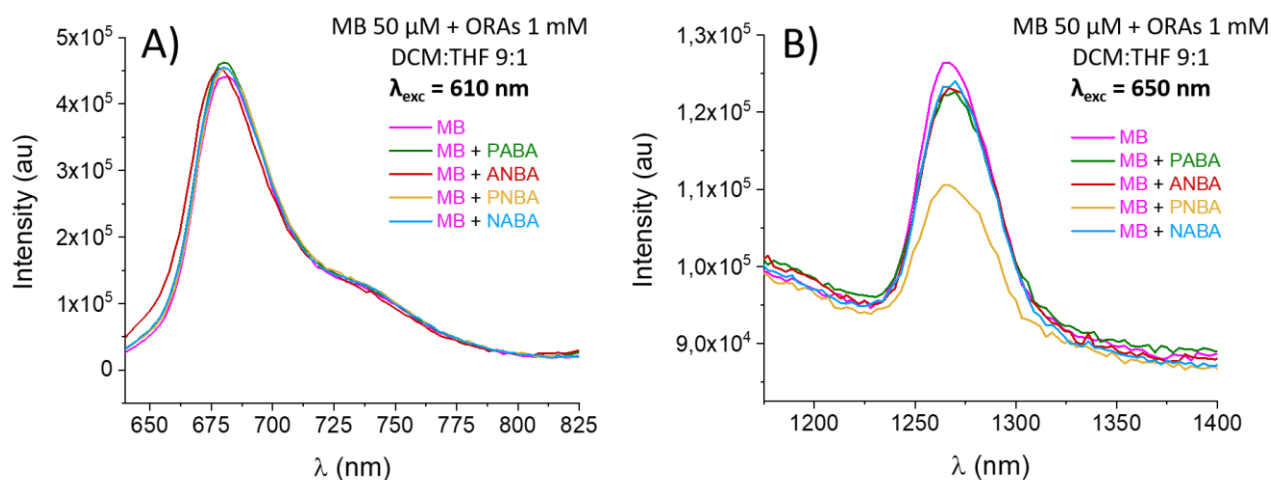


Figure 3: A) Emission spectra of MB (50 μM) alone and in the presence of aromatic substrates (1 mM) with $\lambda_{\text{exc}} = 610$ nm; B) Emission spectra of the $^1\text{O}_2$ produced after irradiation of MB (50 μM) alone and in the presence of the aromatic substrates (1 mM) with $\lambda_{\text{exc}} = 650$ nm. Both experiments were performed in DCM 9:1 THF using a 0.5×0.5 cm cuvette.

The experimental setup for the photooxygenation is described in the **ESI (Figure S5)**. The selected light source was a medium-pressure Hg-lamp (100 W) equipped with a bandpass filter at 577 nm and the samples were irradiated at r.t. under continuous stirring. Air-equilibrated DCM:THF solutions containing MB (50 μM) and the aromatic substrates (1 mM) were irradiated in 1 cm cuvettes and the absorption spectra of the solutions was checked at selected time points in 0.1 cm cuvettes (**Figure 4A**). The relative absorbance values were plotted against time (**Figure 4B**) for the four synthesized substrates and two reference compounds, 9,10-

diphenylanthracene (**DPA**) and anthracene-9,10-diprionic acid (**ADPA**). The same was done for the MB peak at 655 nm and the whole set of absorption spectra is shown in the **ESI (Figure S6)**.

The two anthracene derivatives **PABA** and **ANBA** reacted quantitatively as confirmed by the disappearing of the absorption bands peaking at 375 nm (**PABA**) and 367 nm (**ANBA**) in 30 and 60 minutes, respectively, times comparable to those of **ADPA** (10 min) and **DPA** (20 min) reference compounds. On the contrary, the 1- and 1,4-substituted naphthalenes, **NABA** and **PNBA**, did not show any change in their absorption spectra suggesting no reaction happened under the experimental conditions. We can exclude that the problem could be related to $^1\text{O}_2$ amount produced by MB as it does not depend on the presence of **NABA** and is only halved by **PNBA**. Moreover, these results were corroborated by published data reporting more selective conditions for the conversion of naphthalenes and stabilization of the corresponding endoperoxides.^[15]

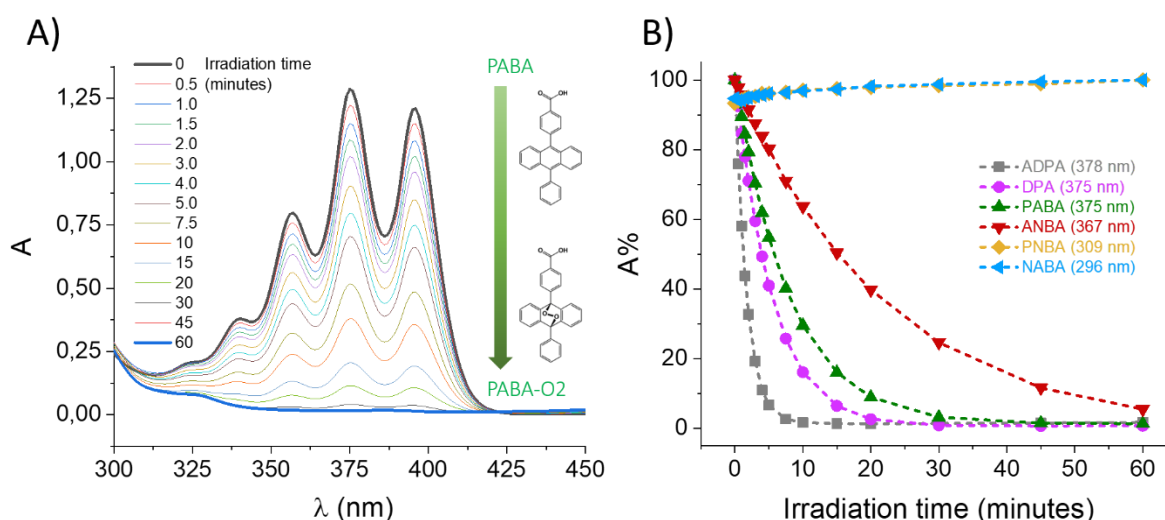


Figure 4: A) Absorption spectra (0.1 cm cuvette) monitoring the photochemical conversion of **PABA** to **PABA-O2** (1 mM) + MB 50 μM in DCM 9:1 THF at r.t.) during irradiation with Hg lamp (interference filter at 577 nm). B) Comparison between the conversion rates of **PABA** (375 nm), **ANBA** (367 nm), **PNBA** (309 nm) and **NABA** (296 nm) with **DPA** (375 nm) and **ADPA** (378 nm) as references.

Overall, the optimized procedure reported here allowed to efficiently synthesize under mild conditions **PABA-O2** and **ANBA-O2**, which were easily purified and isolated by column chromatography with 94% and 84% yields, respectively. Furthermore, the amount of PS was reduced by 50%, the percentage of photochemically unstable THF^[38] was reduced from 33% to 10% and the irradiation setup was more selective thus decreasing energy requirements. The concentration of the starting material was lower, but the batch size was increased, the reaction time reduced and no need of gaseous O_2 atmosphere was requested as atmospheric air pressure and r.t. were applied. Overall, in the same reaction time but under much cheaper conditions, the optimized procedure could deliver > 40% of the product (**PABA-O2**) with respect to the published procedure, making it very suitable for further scale up and future development.

The reaction products **PABA-O2** and **ANBA-O2** were characterized by ESI-MS spectrometry, where the peaks of [**PABA-O2** + H]⁺, [**PABA-O2** + Na]⁺ and [**ANBA-O2** + H]⁺ could be detected (**ESI**). Moreover, ^1H and ^{13}C NMR spectra of the compounds were fully assigned (**ESI**) thanks to two-dimensional ^1H - ^1H COSY, ^1H - ^{13}C HSQC, ^1H -

^{13}C HMBC and 2D ROESY NMR spectroscopy. The formation of the endoperoxide products could be observed first of all from the ^1H NMR spectra (**Figure 5**). The shift of the anthracene-core ^1H resonances (highlighted in green) is highly evident, since the loss of extended aromaticity and planar geometry move the related signals from ≥ 7.4 ppm to ≤ 7.4 ppm. Also the 2D ROESY correlations witnessed the change in geometry upon endoperoxide formation. The planar anthracene moiety of **PABA** induces the aromatic substituents in positions 9 and 10 to remain almost perpendicular to the plane of the central core, hindering the development of dipolar through-space interactions. However, in the case of **PABA-O2**, the anthracene planarity is lost, the geometry of the molecule changed and the protons of rings A and C get closer to the protons of the substituents giving rise to the cross-peaks observed in the 2D spectra. Eventually, the ^{13}C NMR spectra confirmed the **PABA-O2** formation by showing the presence of two signals at *ca.* 84 ppm not present in the spectrum of **PABA**, highlighting the change of hybridization of the aromatic ^{13}C -*sp*² to the non-aromatic ^{13}C -*sp*³ of the atoms bearing the endoperoxide bridge (ring B).

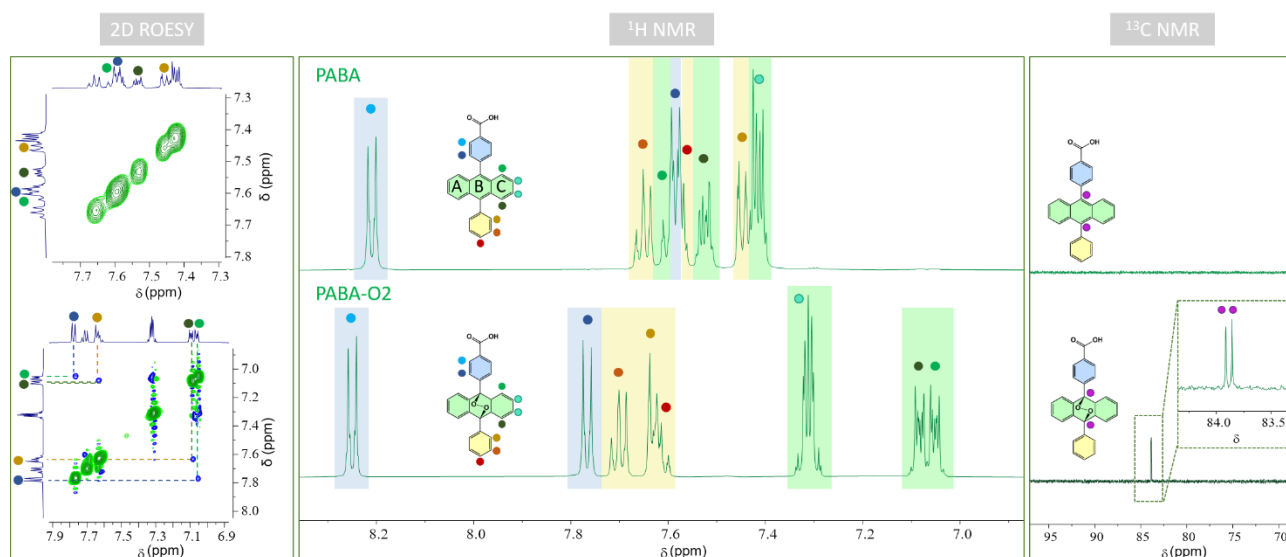


Figure 5: Portions of the 2D ROESY, ^1H and ^{13}C NMR spectra highlighting the main differences between **PABA** and **PABA-O2** (500 MHz, DMSO-d_6). Full spectra are shown in ESI.

The isolated endoperoxides **PABA-O2** and **ANBA-O2** were stable at r.t. as solid powder and in solution for several months. For precaution, they were stored at -20°C for even longer periods.

Solubilisation of aromatic substrates and PS in water: interaction with BCDPS

Stimulated by the envisaged application of ORAs in the treatment of hypoxic conditions, a further challenge was to transfer them in physiologically compatible environment without covalent modification of the ORAs. Highly hydrophilic BCDPS was used as solubilizing matrix for this purpose in aqPBS buffer (pH 7.4).

The solubility of **PABA** and **ANBA** was explored with increasing amounts of BCDPS, up to its maximum solubility in water, 200 mg/mL (**Figure S7**). Phase-solubility diagrams have been recorded for the two compounds, obtaining a curve of the A-type for **PABA** and A_N -type for **ANBA**, assuming in BCDPS a content of monomeric βCyD = 70% w/w (**Figure 6**). The linear A-type profile is indicative of a single complex with 1:1

stoichiometry while the negative deviation from linearity for the A_N -type can have many origins, like the formation of complexes with different stoichiometry, a change in viscosity of the solution, or aggregation of host with its increasing concentration, all explanations reasonable for the polymeric CyD systems.^[39] A change in viscosity can be ruled out, since deviation is observed only for **ANBA**. Considering the dimension of the aromatic molecules, we can envisage the existence of complexes with stoichiometry higher than 1:1, with two CyD molecules complexing one anthracene. While the solubility of the anthracene derivatives alone in water is negligible, the solubility of **ANBA** and **PABA** can increase up to 30 mM and 3 mM, respectively, demonstrating a high efficiency of the polymer to solubilize the synthesized aromatic molecules and leaving room for future developments of these reactions at higher concentrations. Interestingly, the absorption spectra in **Figure S8** show that complexed **ANBA** and **PABA** in aqPBS absorb up to 415 and 425 nm, respectively and their fluorescence lifetime increased significantly with respect to the organic solvent, becoming ≈ 10 ns (**Table S4**), opening new perspectives for further implementation of these molecules as visualizing entities in time-resolved fluorescence techniques. Differently, the fluorescence lifetime of the naphthalene substrates did not increase upon complexation remaining ≤ 2.5 ns.

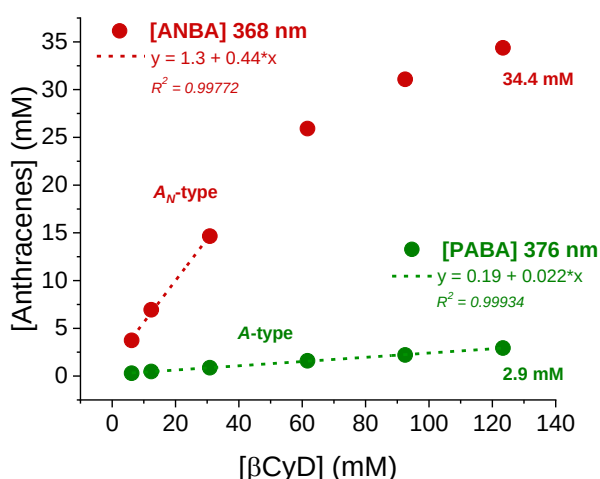


Figure 6: Phase-solubility diagrams of **PABA** and **ANBA** with increasing amounts of BCDPS in aqPBS (pH 7.4). The concentration of the anthracene derivatives is given as function of the estimated amount of β CyD monomer (approx. 70% w/w of BCDPS).

2D ROESY NMR spectra were performed in deuterated PBS (dPBS, pH 7.4) in order to evaluate the type of interaction between the synthesized aromatic substrates and the cavity of 2-hydroxypropyl- β CyD (HP β CyD) monomer, selected as the most plausible reference for BCDPS which present broad ^1H NMR resonances (**Figure S9A**).^[27] **PABA** exhibited solubility issues in the presence of up to 11 equivalents of HP β CyD and only 0.2 mM of the anthracene could be solubilized. This solution resulted in a poor signal-to-noise 2D ROESY NMR spectrum showing weak cross-peaks between the primary side of HP β CyD and the benzoic acid moiety in position 9 of **PABA** (**Figure S9**). Interestingly, no interaction could be observed with the secondary side of HP β CyD, indicating a selective directionality of the complex formed. A much stronger interaction, instead, could be detected in the case of **ANBA**. Up to 5.1 mM of compound was dissolved in dPBS in the presence of

3 equivalents of HP β CyD (**Figure S9**) and the intensity of the corresponding cross-peaks displayed in the 2D ROESY NMR spectrum confirmed a stronger interaction of the benzoic acid substituent with both the primary and the secondary sides of HP β CyD. This suggests a different orientation of the HP β CyD cavity towards the benzoic acid moiety with respect to PABA, suggesting an important role of the guest in explaining the higher apparent solubility described above for the anthracene derivatives.

Eventually, also the interaction of MB with BCDPS in aqPBS was studied. MB aggregates in water at concentration $> 10 \mu\text{M}$ ^[40] and it is known to interact with monomeric β CyD with a $\log K_{1:1} = 3.6$ (buffered water pH 7.2).^[41] A $5 \mu\text{M}$ MB solution was titrated with increasing amounts of BCDPS monitoring both absorption and fluorescence spectra. The obtained data showed a significant change in the MB signals, indicating complexation of MB into the polymeric matrix (**Figure 7**). Fitting both set of spectra with the software ReactLab® and considering a content of monomeric β CyD $\approx 70\%$ w/w, converged with a model of two complexes with 1:1 and 2:1 (β CyD:MB) stoichiometry, displaying $\log K_{1:1} \approx 4.7$ and $\log K_{2:1} \approx 2.1$ for the first and second binding event, respectively (**Figure S10**). This affinity is high considering the rather good solubility of MB in water.

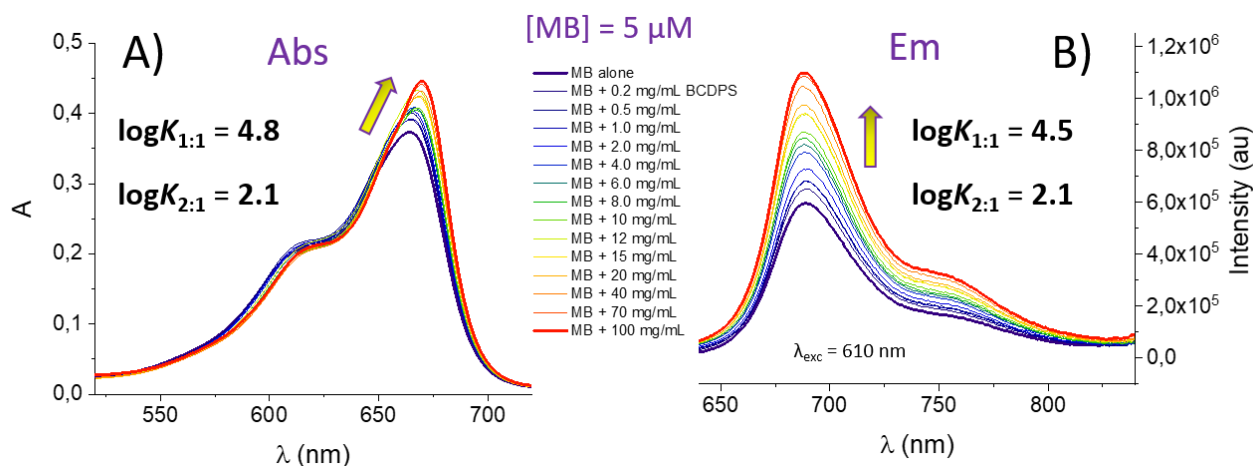


Figure 7: A) Absorption and B) emission ($\lambda_{\text{exc}} = 610 \text{ nm}$) spectra of MB ($5 \mu\text{M}$, 1 cm cuvette) in aqPBS (pH 7.4) with increasing amounts of BCDPS. Values of $\log K$ were obtained for 1:1 and 2:1 (β CyD:MB) complexes by fitting the datasets in ReactLab®.

The occurrence of complexation was also supported by the changes observed in the fluorescence lifetime that, being 0.3 ns for MB alone, becomes biexponential (0.3 ns and 1.2 ns) upon addition of BCDPS. At the end of the titration about 80% of MB resides in the 2:1 complex, so the long lifetime with α value of 0.45 is typical not only for the 2:1 complex but also for the 1:1 complex (**Figure S10D**).

Translation of the photooxygenation reaction in water

Photooxygenation reactions in aqueous solutions were performed with the same setup described above. Initially, dPBS was used to draw benefit from the lifetime of $^1\text{O}_2 = 68 \mu\text{s}$ in D_2O , comparable with the one in DCM ($99 \mu\text{s}$).^[42] Note that the amount of O_2 dissolved in water at ambient pressure is 7 times lower than

DCM and corresponds to 0.3 mM (**Table S1**). All four anthracene and naphthalene substrates were tested, expecting that either the solvent polarity or complexation to the carbohydrate polymeric matrix could change the reactivity of the aromatic molecules^[43] and improve the photooxygenation yields.^[44] To establish conditions similar to those in organic solvent, 50 μM MB was dissolved in 10 mg/mL BCDPS, (MB is 90% complexed as 2:1 and 10% as 1:1 complex), to keep its absorbance values at 577 nm similar to organic solvent. Control experiments irradiating the PS without aromatic substrates suggest an overall enhanced photostability of MB in this solvent upon BCDPS complexation (**Figure S11**).

As expected, PABA dissolved up to ≈ 0.4 mM, while ANBA, PNBA and NABA reached the expected concentration of 1 mM, identical to organic solvent. The photooxygenation was monitored at the same time intervals as for the organic solvent (**Figure S12**) and the set of experiments was completed with the water-soluble reference ADPA (**Figure S13A**). Once again, the naphthalene derivatives PNBA and NABA did not show any reaction, while the anthracene-based PABA and ANBA totally disappeared in ≤ 45 minutes (**Figure 8A**).

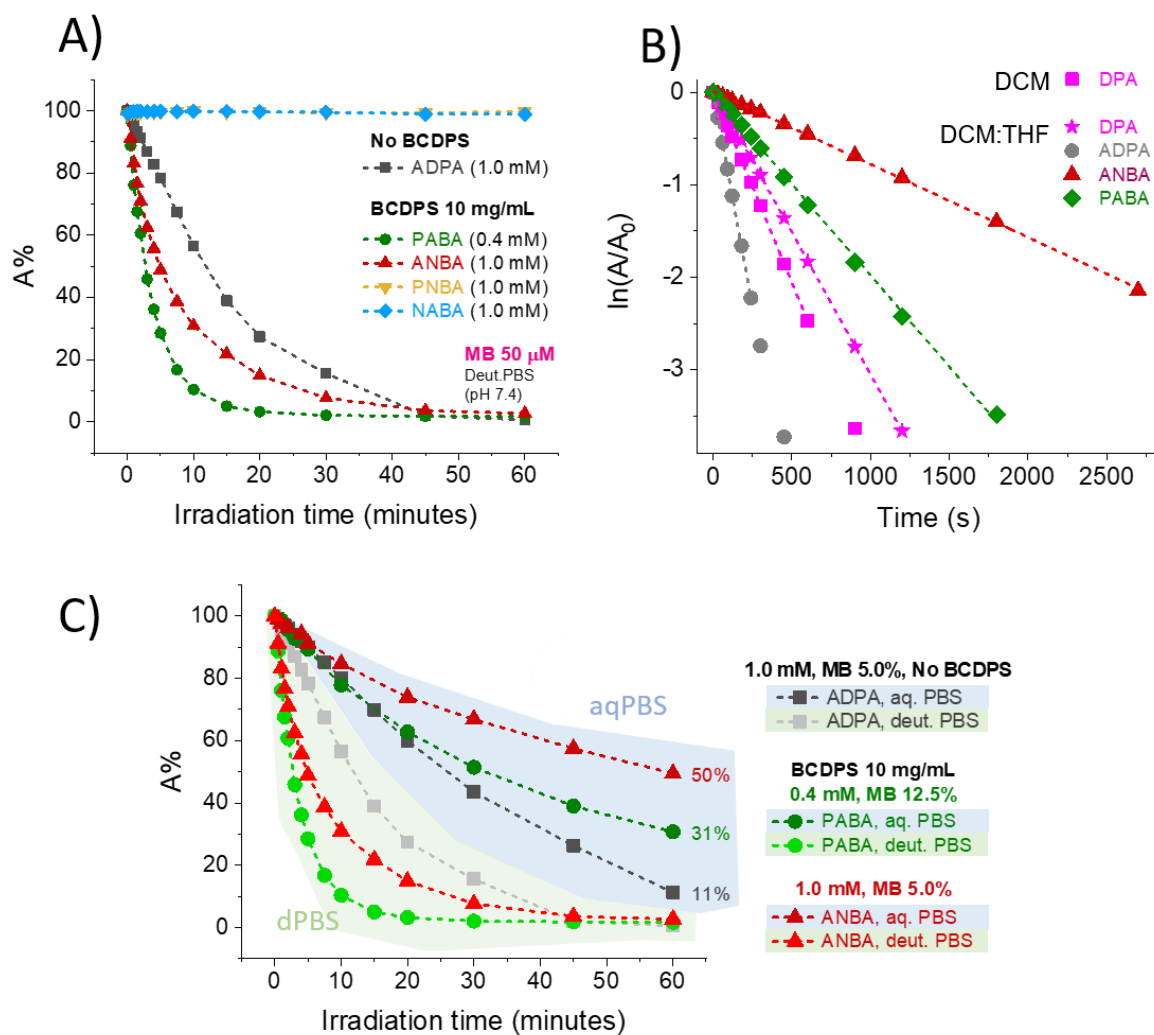


Figure 8: A) Conversion of PABA, ANBA, PNBA and NABA during irradiation with Hg lamp (interference filter at 577 nm) of dPBS solutions containing MB (50 μM) and BCDPS (10 mg/mL). A reference compound (ADPA) was compared in the same solvent without BCDPS. B) Semi logarithmic plots of the photooxygenations of the aromatic substrates in DCM:THF. C) Conversion profiles of ADPA, PABA and ANBA in the same conditions, but different solvent: dPBS vs aqPBS.

Noticeably, MB was not suffering any photobleaching. After extraction in DCM and solvent evaporation, the ^1H NMR spectra of the isolated solids confirmed the formation of the same, fully characterized molecular structures obtained in organic solvent (**Figure S14**). As no starting material as well as no side products could be observed in the spectra, we can conclude the full conversion of the substrates and quantitative formation of the endoperoxides **PABA-O2** and **ANBA-O2**. Moreover, the ^1H NMR spectrum of the aqueous phase in dPBS did not show any aromatic residue thus indicating the possibility to recover and reuse BCDPS as reaction matrix. The photooxygenation reaction was performed also at lower concentration, equal for all substrates (0.2 mM), below the O_2 solubility limit in the solvent and maintaining the same [MB]:[substrate] *ratio* (5.0% n/n), while the rest of the reaction environment (BCDPS 10 mg/mL in dPBS) was kept unaltered (**Figure S15**). When compared to the organic solvent, conversion occurred with similar reaction times for **PABA**, while the performance of **ANBA** conversion resulted to be significantly improved in dPBS, despite the lower O_2 concentration (**Figure S16**). Noticeably, these results evidenced that air-equilibration of the solution was enough to provide the needed amount of O_2 for the photooxygenation without supplying any extra flow of gaseous O_2 in the reaction mixture.

Comparing the reaction kinetics by plotting $\ln(A/A_0)$ vs time is not straightforward (**Figure 8B**). Even considering that also MB might quench $^1\text{O}_2$, its constant contribution, being present in 50 μM , is not relevant considering a quenching rate constant $k = 5 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$ in chlorinated solvent.^[45] The reaction rate constants of the anthracenes with $^1\text{O}_2$ in DCM:THF 9:1 are in line with already published data for **DPA** in plain DCM and further show that monosubstituted anthracene reacts slower compared to the disubstituted molecule (**Table S5, Figure S17**). In aqueous environment, linearity of $\ln(A/A_0)$ vs time is observed for much lower conversion ratios. The reported rate constants in H_2O and D_2O for the reaction of MB with $^1\text{O}_2$ are $3 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$ and $4 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$, respectively. Again, MB in 50 μM solution is not significantly contributing, considering k_d values of $1.5 \times 10^4 \text{ s}^{-1}$ and $2.5 \times 10^5 \text{ s}^{-1}$ in D_2O and H_2O . So, lack in linearity is likely due to much high rate constants in water for the reaction of the anthracenes with $^1\text{O}_2$. Literature reports a value of $1 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$ for ADPA in water,* which makes the quenching of $^1\text{O}_2$ by the aromatic substrate more important than its physical quenching.^[45,46]

Inclusion of both the anthracene and MB in the polymer brings them at close proximity and this may positively impact the endoperoxide formation not requiring any diffusion-controlled encounter of the anthracene and $^1\text{O}_2$ the latter being generated close to the scavenger. This may also explain why it is not possible to observe the $^1\text{O}_2$ phosphorescence not even in the deuterated solvent (**Figure S18**).

Eventually, the last challenge was the translation of the reaction in non-deuterated PBS (**Figure S19**), knowing that the lifetime of $^1\text{O}_2$ in H_2O is much shorter (3 μs) than in deuterated and organic solvents. In 60 minutes,

* In the case of the reference ADPA, where no BCDPS is present, MB is partially aggregated. This behavior is known to inhibit $^1\text{O}_2$ formation and the number of photons absorbed differs from the other solutions in spite of the equimolar conditions. Indeed, the polymer reduces MB aggregation.

the synthesized anthracene derivatives dissolved in the polymeric reaction vessel were converted into their endoperoxide-analogues up to 69% and 50% for **PABA** and **ANBA**, respectively (**Figure 8C**). This is the first time that hydrophobic anthracene-derivatives are converted into their endoperoxide-forms in non-deuterated aqueous environment and such results evidence the pivotal role of BCDPS as suitable matrix for the green homogeneous photocatalytic oxygenation of organic molecules in plain water.

Cycloreversion

The feasibility of the photooxygenation reaction in aqueous environment of the anthracene substrates after complexation in BCDPS allowed to indirectly evaluate the solubility and stability of the formed endoperoxides **PABA-O2** and **ANBA-O2** under these conditions. Precipitation of solid material was never observed after complete conversion of the substrates and the solutions were stable at r.t. for weeks. Hence, aiming to develop a multipurpose polymeric platform to be exploited both for the synthesis and delivery of ORAs, a preliminary evaluation of the cycloreversion step of **PABA-O2** and **ANBA-O2** was performed.

First of all, two triggers were explored in organic solvent ($\text{DMSO-}d_6$): heating at 90 °C and irradiation in the UV region, monitoring the solutions with UV-Vis and ^1H NMR spectroscopies (**Figure S20**). As expected, **PABA-O2** showed a smooth reversion to **PABA** upon thermolysis, matching published data.^[33] In the UV-Vis spectra the characteristic band of the anthracene core was firstly formed and then constantly increasing during continuous heating at 90 °C (**Figure 10A**). After 8 h, around 50% of **PABA-O2** was converted.

Monitoring the most deshielded protons in the ^1H NMR spectra (**Figure 10B**), it could be observed that $\sim 80\%$ of **PABA-O2** was converted to **PABA** after 24 h of heating at 90 °C. The lack of any observable degradation product suggested that only one clean cycloreversion process was obtained in these conditions. Photolysis of **PABA-O2** underscored thermolysis. The UV-Vis absorption spectra showed the appearance of the typical anthracene band, which however reached a plateau after about 3 h of irradiation. As already documented,^[10,17] irradiation at 265 nm probably launched the homolytic cleavage of the endoperoxide bridge, but it also gave rise to a series of side-reactions as soon as a very little amount of **PABA** was regenerated, absorbing the 265 nm light itself thus triggering unwanted side processes, like the heterolytic cleavage of the O-O bond. In the case of **ANBA-O2**, neither thermolysis nor photolysis resulted in the regeneration of the parent compound (**Figure 10C**). With this substrate, the UV-Vis absorption spectra resulted to be substantially unchanged, while the ^1H NMR spectra highlighted the formation of many different species in solution upon continuous heating at 90 °C and very little amount of **ANBA** (**Figure S20**). These data were corroborated by ESI-MS, which showed peaks corresponding to hydroxylated products for the endoperoxide **ANBA-O2** (**ESI**), probably generated during ionization of the molecule.

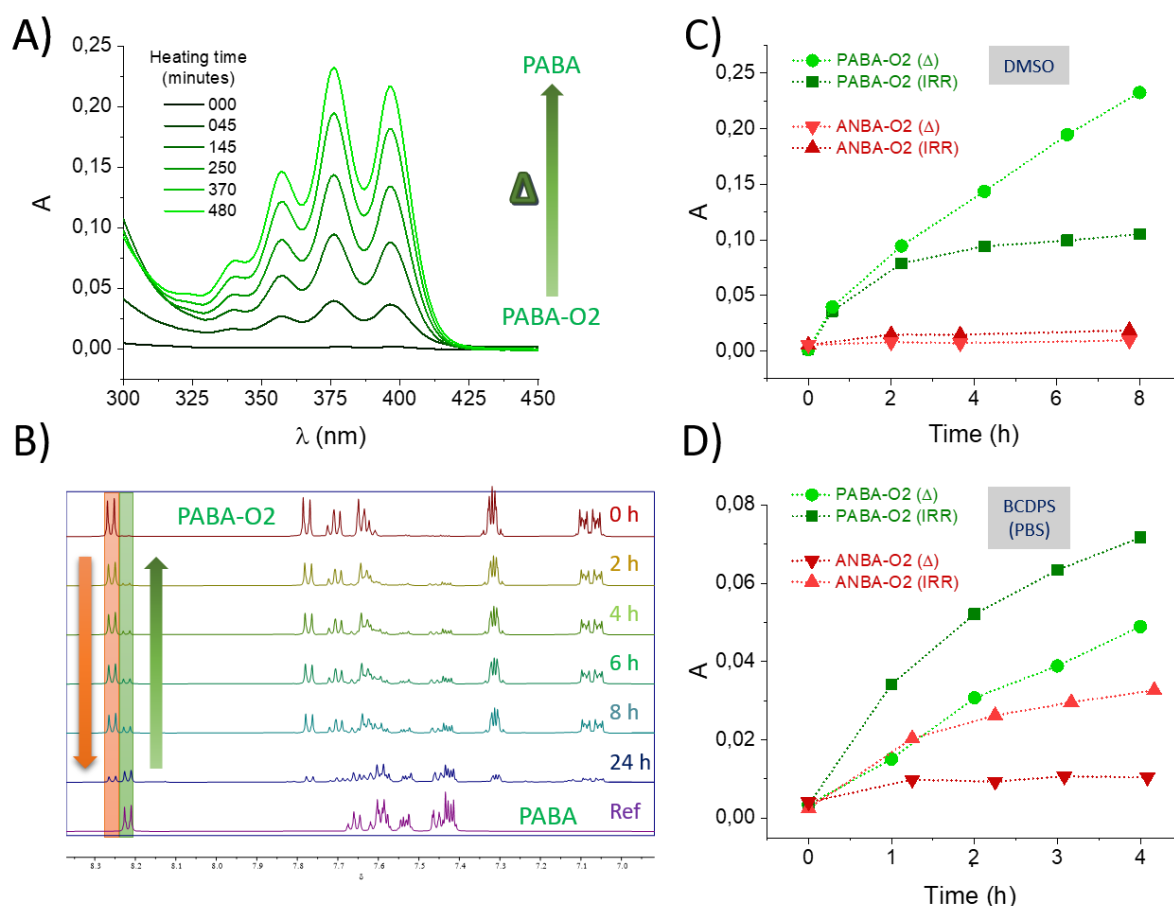


Figure 9: A) Absorption (0.2 mM) and B) ¹H NMR (53 mM, 500 MHz) spectra of PABA-O2 during heating at 90 °C in DMSO-d₆ at different intervals of time. Comparison of the thermolysis (Δ) or photolysis (IRR) of 0.2 mM PABA-O2 and ANBA-O2 reverting to the corresponding parent compounds, PABA and ANBA, respectively, in C) DMSO-d₆ and D) aqPBS in the presence of BCDPS (10 mg/mL).

The same cycloreversion procedures were performed on both PABA-O2 and ANBA-O2 in aqPBS, after complexation with BCDPS (Figure S21). Once again, thermolysis at 90 °C of PABA-O2 showed a linear increase, while irradiation at 265 nm gave only partial photolysis resulting in a flattening of the curve (Figure 10D). For ANBA-O2, the performance was again inferior with respect to PABA-O2. However, even though heating did not start any thermolytic process, irradiation at 260 nm induced at least a partial formation of ANBA, which was not observed in the previously described conditions.

These preliminary results were obtained in non-biocompatible conditions; however, the data encourage the feasibility of cycloreversion reaction of endoperoxides in physiological environment.

Conclusions

Straightforward synthesis of aromatic substrates to be used for the preparation of oxygen releasing agents (ORAs) was achieved and significantly optimized compared to published procedures. The synthesized substrates, two pairs of di- and mono-substituted anthracene (PABA, ANBA) and naphthalene (PNBA, NABA) derivatives were subject of photooxygenation in the presence of methylene blue (MB) as low-cost PS in a mixture of organic solvents, reaching the goal of endoperoxidation only for the anthracenes. The stable

oxygenated **PABA-O2** and **ANBA-O2** were prepared using a selective irradiation setup, thus achieving mild conditions and improved yields ($\geq 84\%$). Full characterization by UV-Vis, ESI-MS and NMR analysis confirmed the expected structures.

By means of a biocompatible cyclodextrin-based polymeric matrix (BCDPS), the aromatic molecules were effectively dissolved in the mM range in aqueous environment. After co-encapsulation of catalytic amounts of MB into BCDPS and following the same protocol used in the case of organic solvents, the anthracenes **ANBA** and **PABA** were photochemically transformed in their endoperoxide analogues in PBS buffer (pH 7.4) of both D₂O and H₂O. This latter represents an extraordinary result as ¹O₂ has a much shorter lifetime in this solvent. Naphthalene-substrates, instead, were not suitable for photooxygenation neither in aqueous solvent. Cycloreversion of **PABA-O2** was achieved upon thermolysis in both organic and aqueous environments even though in conditions not yet compatible with *in vivo* applications. Interestingly, the anthracene derivatives complexed to BCDPS absorb up to 415 nm, making them suitable fluorescent probe in imaging applications. Moreover, their longer lifetimes enabling their discrimination from biological species by means of fluorescence lifetime imaging, afford an extra tool to monitor the O₂ release in complex physiological environments.

Hence, BCDPS was shown to act both as solubilizing agent and reaction vessel enabling the photooxygenation in buffered water of hydrophobic substrates with an economical setup, comparable concentrations and improved reaction times with respect to the organic solvent. These results are of great importance for future exploitations in green synthesis and have the potential to launch BCDPS as affordable hydrophilic platform for the development of more sustainable homogeneous photocatalytic protocols. Exploiting the versatility of the polymer to simultaneously solubilize different hydrophobic substrates bringing them in close proximity, also other types of reactions can be envisaged on hydrophobic molecules in aqueous environment. Moreover, once optimized the cycloreversion conditions, the use of a unique inexpensive polymeric matrix to solubilize, synthesize and deliver ORAs might definitively push the fast developments of new treatments for diseases displaying hypoxic environments.

Acknowledgements



This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No. 894942 (HypoCyclo) and the Young Investigator Training Program (YITP) 2019.

Open Access Funding provided by Consiglio Nazionale delle Ricerche within the CRUI-CARE Agreement.

Conflict of Interest

The authors declare no competing financial interests.

Data Availability Statement

The data that support the findings of this study are available in the Electronic Supporting Information (ESI) of this article.

References

- [1] X. Li, N. Kwon, T. Guo, Z. Liu, J. Yoon, *Angew. Chemie Int. Ed.* **2018**, *57*, 11522–11531.
- [2] S. R. McKeown, *Br. J. Radiol.* **2014**, *87*, 1–12.
- [3] A. C. Gonçalves, E. Richiardone, J. Jorge, B. Polónia, C. P. R. Xavier, I. C. Salaroglio, C. Riganti, M. H. Vasconcelos, C. Corbet, A. B. Sarmento-Ribeiro, *Drug Resist. Updat.* **2021**, *59*, DOI 10.1016/j.drug.2021.100797.
- [4] F. Marcucci, C. Rumio, *Neoplasia (United States)* **2021**, *23*, 234–245.
- [5] B. Schaible, C. T. Taylor, K. Schaffer, *Antimicrob. Agents Chemother.* **2012**, *56*, 2114–2118.
- [6] C. H. Kowalski, K. A. Morelli, D. Schultz, C. D. Nadell, R. A. Cramer, *Proc. Natl. Acad. Sci. U. S. A.* **2020**, *117*, 22473–22483.
- [7] F. Adhami, G. Liao, Y. M. Morozov, A. Schloemer, V. J. Schmithorst, J. N. Lorenz, R. S. Dunn, C. V. Vorhees, M. Wills-Karp, J. L. Degen, R. J. Davis, N. Mizushima, P. Rakic, B. J. Dardzinski, S. K. Holland, F. R. Sharp, C. Y. Kuan, *Am. J. Pathol.* **2006**, *169*, 566–583.
- [8] G. Loor, P. T. Schumacker, *Cell Death Differ.* **2008**, *15*, 686–690.
- [9] W. Boulefour, E. Rowinski, S. Louati, S. Sotton, A. S. Wozny, P. Moreno-Acosta, B. Mery, C. Rodriguez-Lafrasse, N. Magne, *Med. Sci. Monit.* **2021**, *27*, 1–7.
- [10] J.-M. Aubry, C. Pierlot, J. Rigaudy, R. Schmidt, *Acc. Chem. Res.* **2003**, *36*, 668–675.
- [11] E. L. Clennan, *Photochem. Photobiol.* **2022**, 1–17.
- [12] W. FUDICKAR, T. LINKER, in *PATAI'S Chem. Funct. Groups* (Ed.: L. John Wiley & Sons), John Wiley & Sons, Ltd, Chichester, UK, **2014**, pp. 1–66.
- [13] W. Fudickar, T. Linker, *J. Org. Chem.* **2017**, *82*, 9258–9262.
- [14] W. Fudickar, T. Linker, *Chem. - A Eur. J.* **2011**, *17*, 13661–13664.
- [15] W. Fudickar, T. Linker, *ChemPhotoChem* **2021**, *5*, 1004–1008.
- [16] M. Klaper, T. Linker, *Chem. - A Eur. J.* **2015**, *21*, 8569–8577.
- [17] W. Fudickar, T. Linker, *ChemPhotoChem* **2018**, *2*, 548–558.

- [18] M. Klaper, P. Wessig, T. Linker, *Chem. Commun.* **2016**, 52, 1210–1213.
- [19] W. Fudickar, T. Linker, *Angew. Chemie Int. Ed.* **2018**, 57, 12971–12975.
- [20] P. Jansook, N. Ogawa, T. Loftsson, *Int. J. Pharm.* **2018**, 535, 272–284.
- [21] W. Zhao, Q. Zhong, *J. Incl. Phenom. Macrocycl. Chem.* **2012**, 72, 1–14.
- [22] F. Ortega-Caballero, C. Rousseau, B. Christensen, T. E. Petersen, M. Bols, *J. Am. Chem. Soc.* **2005**, 127, 3238–3239.
- [23] S. Tilloy, H. Bricout, E. Monflier, *Green Chem.* **2002**, 4, 188–193.
- [24] G. Floresta, C. Talotta, C. Gaeta, M. De Rosa, U. Chiacchio, P. Neri, A. Rescifina, *J. Org. Chem.* **2017**, 82, 4631–4639.
- [25] M. Bonchio, T. Carofiglio, M. Carraro, R. Fornasier, U. Tonellato, *Org. Lett.* **2002**, 4, 4635–4637.
- [26] X. F. Chen, D. K. P. Ng, *Chem. Commun.* **2021**, 57, 3567–3570.
- [27] L. Slavětínská, J. Mosinger, P. Kubát, *J. Photochem. Photobiol. A Chem.* **2008**, 195, 1–9.
- [28] B. A. Davis, J. A. Bennett, J. Genzer, K. Efimenko, M. Abolhasani, *ACS Sustain. Chem. Eng.* **2022**, DOI 10.1021/acssuschemeng.2c05467.
- [29] T. Huang, G. Sheng, P. Manchanda, A. H. Emwas, Z. Lai, S. P. Nunes, K. V. Peinemann, *Sci. Adv.* **2019**, 5, 1–11.
- [30] M. Agnes, E. Pancani, M. Malanga, E. Fenyvesi, I. Manet, *Macromol. Biosci.* **2022**, 22, e2200090.
- [31] C. F. R. A. C. Lima, L. R. Gomes, L. M. N. B. F. Santos, J. N. Low, *Acta Crystallogr. Sect. E Struct. Reports Online* **2009**, 65, 148–155.
- [32] H. Jintoku, M. T. Kao, A. Del Guerzo, Y. Yoshigashima, T. Masunaga, M. Takafuji, H. Ihara, *J. Mater. Chem. C* **2015**, 3, 5970–5975.
- [33] S. Kolemen, T. Ozdemir, D. Lee, G. M. Kim, T. Karatas, J. Yoon, E. U. Akkaya, *Angew. Chemie Int. Ed.* **2016**, 55, 3606–3610.
- [34] M. Agnes, A. Arabi, M. Caricato, A. Nitti, D. Dondi, K. Yannakopoulou, M. Patrini, D. Pasini, *Chempluschem* **2021**, 86, 270–274.
- [35] F. Wilkinson, W. P. Helman, A. B. Ross, *J. Phys. Chem. Ref. Data* **1993**, 22, 113–262.
- [36] R. E. Ballard, C. H. Park, *J. Chem. Soc. A Inorganic, Phys. Theor.* **1970**, 1340.
- [37] S. L. Fornili, G. Sgroi, V. Izzo, *J. Chem. Soc. Faraday Trans. 1 Phys. Chem. Condens. Phases* **1981**, 77, 3049–3053.
- [38] A. Sagadevan, K. C. Hwang, M.-D. Su, *Nat. Commun.* **2017**, 8, 1812.
- [39] M. E. Brewster, T. Loftsson, *Adv. Drug Deliv. Rev.* **2007**, 59, 645–666.
- [40] A. Fernandez-Perez, G. Marban, *ACS Omega* **2020**, 5, 29801–29815.

- [41] G. C. Zhao, J. J. Zhu, J. J. Zhang, H. Y. Chen, *Anal. Chim. Acta* **1999**, 394, 337–344.
- [42] C. Schweitzer, R. Schmidt, *Chem. Rev.* **2003**, 103, 1685–1757.
- [43] J.-M. Aubry, B. Mandard-Cazin, M. Rougee, R. V. Bensasson, *J. Am. Chem. Soc.* **1995**, 117, 9159–9164.
- [44] M. Bauch, W. Fudickar, T. Linker, *Molecules* **2021**, 26, 804.
- [45] F. Wilkinson, W. P. Helman, A. B. Ross, *J. Phys. Chem. Ref. Data* **1995**, 24, 663–677.
- [46] M. Hoebeke, X. Damoiseau, *Photochem. Photobiol. Sci.* **2002**, 1, 283–287.